Multi-Model Deep Learning for Dermoscopic Skin Lesion Classification on HAM10000

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Abstract

Automated classification of skin lesions from dermoscopic images is clinically important and technically challenging, owing to fine inter-class visual similarity and severe class imbalance. We present a comprehensive deep learning pipeline trained on the HAM10000 benchmark (10,015 dermoscopic images, 7 classes). Five architectures of increasing complexity are designed and evaluated: a baseline custom CNN (HAM-CNN), an enhanced CNN with Feature Pyramid Network (FPN), Generalised Mean pooling, stochastic depth, and Sharpness-Aware Minimisation; a dual-branch Double CNN with auxiliary reconstruction; a fine-tuned DINOv2-Small Vision Transformer; and a fine-tuned EfficientNetB3. A weighted ensemble whose coefficients are optimised via Differential Evolution achieves **86.1%** accuracy, **0.807** macro-F1, and **0.966** AUC-ROC on the held-out test set, outperforming all individual models. An interactive Streamlit web application provides inference with Grad-CAM saliency overlays.

Keywords: skin lesion classification, dermoscopy, CNN, Vision Transformer, DINOv2, EfficientNet, Focal Loss, ensemble, HAM10000.

1 Introduction

Skin cancer is among the most prevalent cancers globally, with early and accurate diagnosis being critical for patient outcomes. Dermoscopy substantially improves diagnostic accuracy, yet access to expert dermatologists is unevenly distributed. Automated computer-aided diagnosis systems can serve as decision-support tools, particularly in resource-limited settings.

The HAM10000 dataset [1] contains 10,015 dermoscopic images spanning seven lesion types and poses two key challenges: (i) *fine-grained visual similarity* between malignant and benign lesions; and (ii) *severe class imbalance*, with melanocytic nevi comprising ≈67% of samples. We systematically compare five architectures and combine them into an optimised ensemble, ablating the contribution of each design choice.

2 Related Work

Esteva et al. [2] showed that CNNs trained on large clinical image collections can achieve dermatologist-level accuracy. The ISIC challenge series [3] established standardised benchmarks on HAM10000, with top entries relying on ensembles and heavy augmentation. EfficientNet [4] introduced compound scaling for efficient CNN design. Vision Transformers (ViT) [5] achieve competitive results when pretrained at scale. DINOv2 [6] extends self-supervised ViT pretraining, producing transferable features for fine-grained tasks. Focal Loss [7] and SAM [8] address class imbalance and loss-landscape sharpness, respectively. FPNs [9] aggregate multi-scale features and benefit fine-grained recognition.

3 Dataset and Preprocessing

3.1 Dataset

HAM10000 provides 10,015 JPEG dermoscopic images across seven classes: actinic keratosis (akiec), basal cell carcinoma (bcc), benign keratosis (bkl), dermatofibroma (df), melanoma (mel), melanocytic nevi (nv), and vascular lesion (vasc). Each image carries a patient identifier used to prevent data leakage.

3.2 Preprocessing and Splitting

Images are resized to 224×224 and normalised using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). Outlier images are removed prior to splitting. Patient-disjoint splits are generated with `GroupShuffleSplit`: train (6,800), validation (1,450), test (1,431).

3.3 Class Balancing

Imbalance is mitigated via: (i) **WeightedRandomSampler**, producing a near-uniform per-epoch distribution; and (ii) **class-weighted Focal Loss**, down-weighting easy majority-class examples.

3.4 Data Augmentation

Training uses a dermatoscopy-motivated Albumentations [10] pipeline: geometric transforms (horizontal/vertical flip, RandomRotate90, ElasticTransform, GridDistortion, OpticalDistortion), photometric transforms (CLAHE, RandomShadow, ColorJitter), CoarseDropout, MixUp ($\alpha = 0.4$), and CutMix ($p = 0.5$). Validation and test images receive only resize and normalisation.

4 Methods

4.1 Baseline CNN (HAMCNN)

A five-stage network with channel widths $[32, 64, 128, 256, 384]$, each stage comprising residual blocks followed by 2×2 max-pooling. The head applies global average pooling, a 256-unit hidden layer, dropout ($p = 0.4$), and a 7-class linear output.

4.2 Enhanced CNN

The baseline is augmented with: (1) **FPN** fusing stages 3–5 to a 128-channel space; (2) **GeM pooling** with learnable exponent ($p_0 = 2.0$); and (3) **Stochastic Depth** ($p_{\text{drop}} = 0.15$). Trained with SAM and AMP.

4.3 Double CNN

A shared stem (stages 1–2, $32 \rightarrow 64$ channels) feeds: a **classification branch** (deeper stages with FPN and GeM) and a **reconstruction branch** (transposed-convolution decoder to pixel space). The training objective is:

$$\mathcal{L} = \mathcal{L}_{\text{focal}} + 0.15 \cdot \mathcal{L}_{\text{recon}} \tag{1}$$

The auxiliary task prevents the stem from discarding low-level texture information relevant to lesion characterisation.

4.4 Vision Transformer — DINOv2-Small

`vit_small_patch14_dinov2` from `timm`, fine-tuned after a 5-epoch backbone freeze. Differential LRs: backbone 2×10^{-6}, head 5×10^{-4}. Head: two-layer MLP (hidden dim 512, dropout $p = 0.3$).

4.5 EfficientNetB3

`efficientnet_b3` pretrained on ImageNet-1k, fine-tuned after a 5-epoch freeze. Differential LRs: backbone 5×10^{-5}, head 5×10^{-4}. Head: Dropout(0.4) $\rightarrow$ Linear$(1536, 7)$.

4.6 Weighted Ensemble

Final predictions combine per-class softmax probabilities:

$$\hat{p} = \sum_{m=1}^{5} w_m \cdot \text{softmax}(\mathbf{z}_m) \tag{2}$$

Weights are optimised on the validation set via Differential Evolution [11] (18,200 evaluations, maximising weighted recall, converged successfully). The resulting weights are shown in Table 1.

4.7 Training Protocol

All models share the hyperparameters in Table 2.

Table 1: DE-optimised ensemble weights.

Model	Weight
HAMCNN	0.0297
Enhanced CNN	0.0297
Double CNN	0.1386
ViT-DINOv2	0.6040
EfficientNetB3	0.1980

4.8 Test-Time Augmentation

Two strategies are applied at inference: (1) **Deterministic TTA** (4 passes: original, H-flip, V-flip, both); and (2) **Stochastic TTA** (16 Monte-Carlo dropout passes with training augmentations).

4.9 Grad-CAM Visualisation

Grad-CAM [12] is applied to EfficientNetB3 to identify discriminative spatial regions per predicted class.

5 Experiments and Results

5.1 Metrics

Evaluation on the held-out test set ($n = 1{,}431$) uses: accuracy, macro-averaged F1 (macro-F1), macro-averaged AUC-ROC (one-vs-rest), and Expected Calibration Error (ECE; lower is better).

5.2 Training Dynamics

Figure 1 shows training and validation curves for all five models. The ViT exhibits a characteristic loss spike at epoch 5 when the backbone is unfrozen (red dashed line), followed by rapid convergence. EfficientNetB3 achieves the second-lowest validation loss. The three custom CNNs plateau at substantially higher loss, reflecting limited capacity relative to pretrained models. The ECE chart confirms that ViT achieves the best calibration (ECE = 0.163).

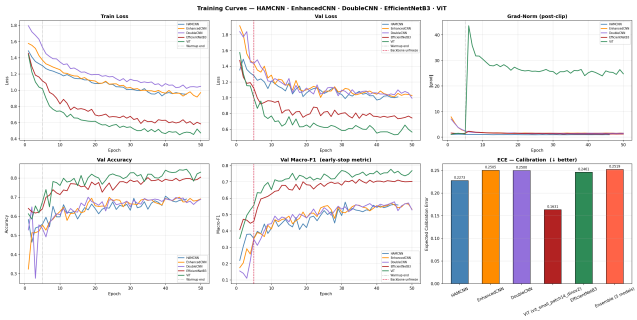

Figure 1: Training dynamics for all five models. *Top row*: train loss, val loss, gradient norm (post-clip). *Bottom row*: val accuracy, val macro-F1 (early-stop metric), ECE. Dotted line: warmup end. Dashed red line: ViT backbone unfreeze.

5.3 Quantitative Results

Quantitative results are reported in Table 3. The ViT-DINOv2 is the strongest single model (Acc=0.853, Macro-F1=0.790, AUC=0.953), reflecting the power of self-supervised pretraining. EfficientNetB3 achieves the highest single-model AUC (0.962). The ensemble improves upon all single models on accuracy and AUC, and achieves the best macro-F1 (0.807), confirming the complementarity of the five architectures.

5.4 Per-class Performance

Figure 2 shows per-class F1 scores. Melanoma (mel) is the hardest class for all models due to visual overlap with melanocytic nevi. Vascular lesion (vasc) and basal cell carcinoma (bcc) are consistently well-classified. The ensemble surpasses all individual models on every class, reaching F1 of 0.91 for vasc and 0.87 for bcc.

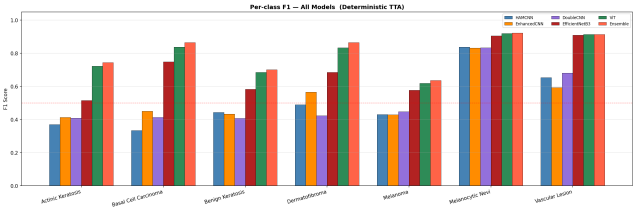

Figure 2: Per-class F1 scores for all models and the ensemble (deterministic TTA, $n = 4$). The red dashed line marks F1 = 0.5. The ensemble (coral bars) consistently dominates all individual models.

5.5 Confusion Matrices

Figure 3 shows normalised confusion matrices. Custom CNNs exhibit systematic mel→nv confusion. ViT and EfficientNetB3 substantially reduce off-diagonal entries, and the ensemble further reinforces diagonal dominance across all seven classes.

5.6 Per-class Report

The full ensemble classification report is given in Table 4.

6 Web Application

The Streamlit application (`app.py`) accepts a user-uploaded image with interactive cropping and returns: (i) predicted class and top-3 class probabilities as an interactive Plotly bar chart; and (ii) a Grad-CAM overlay. Inference uses the ensemble checkpoint on the available device (GPU or CPU).

7 Limitations

- No out-of-distribution detection: the model yields uncalibrated probabilities for non-dermoscopic inputs.
- HAM10000 originates from a single clinical centre; domain shift may degrade performance on images from other dermoscopes.
- This project is intended for *educational purposes only* and must not be used for clinical decision-making.

8 Conclusion

We presented a multi-model deep learning pipeline for skin lesion classification on HAM10000. Five architectures were trained with focal loss, WeightedRandomSampler, and dermatoscopy-specific augmentation, then combined into an ensemble with Differential Evolution-optimised weights. The ensemble achieves 86.1% accuracy, 0.807 macro-F1, and 0.966 AUC-ROC. Future work will investigate uncertainty quantification, OOD detection, and cross-domain fine-tuning.

References

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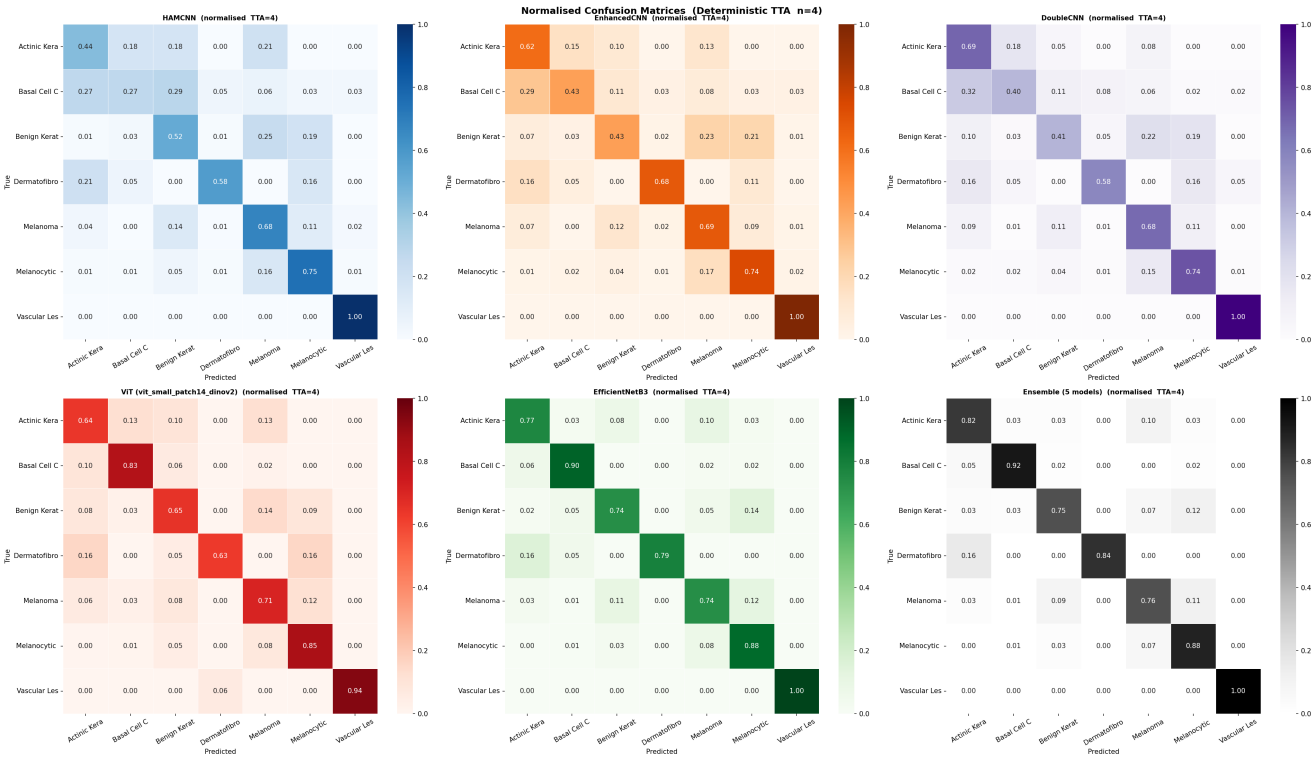

Figure 3: Normalised confusion matrices for all five models and the ensemble (deterministic TTA, $n = 4$). Rows: true class; columns: predicted class. The ensemble (bottom-right) achieves the strongest diagonal dominance across all seven categories.

Table 2: Common training hyperparameters shared by all models.

Hyperparameter	**Value**
Image size	224×224
Batch size	32
Max epochs	50
Early stopping patience	15 epochs (metric: val macro-F1)
Loss function	Focal Loss ($\gamma = 1.5$) + label smoothing ($\varepsilon = 0.05$)
LR schedule	Linear warmup (5 ep.) $\rightarrow$ Cosine annealing
Gradient clipping	$\|\nabla\| \leq 2.0$
Mixed precision (AMP)	✓ (all models)
Seed	42
Hardware	NVIDIA RTX 4050 Laptop GPU (6 GB VRAM)
Framework	PyTorch 2.5.1 + CUDA 12.1

Table 3: Test-set performance of all models and the ensemble (deterministic TTA, $n = 4$). **Bold**: best per column. ECE: lower is better; all other metrics: higher is better.

Model	**Accuracy**	**Macro F1**	**AUC-ROC**	**ECE**	**Parameters**	**TTA**
HAMCNN	0.6953	0.5081	0.9278	0.2273	7,075,239	4
Enhanced CNN	0.6946	0.5310	0.9344	0.2505	7,307,508	4
Double CNN	0.6939	0.5163	0.9305	0.2500	7,348,615	4
ViT-DINOv2-Small	0.8532	0.7903	0.9531	**0.1631**	22,257,159	4
EfficientNetB3	0.8134	0.7036	0.9620	0.2461	11,489,327	4
Ensemble (5 models)	**0.8609**	**0.8070**	**0.9657**	0.2519	—	4

Table 4: Per-class classification report for the weighted ensemble on the test set (TTA $n = 4$).

Class	**Precision**	**Recall**	**F1-score**	**Support**
Actinic Keratosis	0.68	0.82	0.74	39
Basal Cell Carcinoma	0.82	0.92	0.87	63
Benign Keratosis	0.65	0.75	0.70	118
Dermatofibroma	0.89	0.84	0.86	19
Melanoma	0.55	0.76	0.64	140
Melanocytic Nevi	0.97	0.88	0.92	1,036
Vascular Lesion	0.84	1.00	0.91	16
Macro avg	0.77	0.85	0.81	1,431
Weighted avg	0.88	0.86	0.87	1,431
Accuracy			**0.86**	1,431